

Using the Logistic Equation to Model Population Growth

What Happens to the Size of a Population When It Overshoots Its Carrying Capacity? In the logistic population growth model, the per capita rate of population increase approaches zero as the population size (N) approaches the carrying capacity (K). Under some conditions, however, a population in the laboratory or the field can overshoot K , at least temporarily. If food becomes limiting to a population, for instance, there may be a delay before reproduction declines, and N may briefly exceed K . In this exercise, you will use the logistic equation to model the growth of the hypothetical population in Table 53.2 when $N > K$.

INTERPRET THE DATA

1. Assuming that $r = 1.0$ and $K = 1,500$, calculate the population growth rate for four cases where population size (N) is greater than carrying capacity (K): $N = 1,510, 1,600, 1,750$ and $2,000$ individuals. To do this, first write the equation for population growth rate given in Table 53.2. Plug in the values for each of the four cases, starting with $N = 1,510$, and solve the equation for each one. Which population size has the highest growth rate?

► *Daphnia*



2. If r is doubled, predict how the population growth rates will change for the four population sizes given in question 1. Now calculate the population growth rate for the same four cases, this time assuming that $r = 2.0$ (and with K still = 1,500).
3. Now let's see how the growth of a real-world population of *Daphnia* corresponds to this model. At what times in Figure 53.10b is the *Daphnia* population changing in ways that correspond to the values you calculated? Hypothesize why the population drops below the carrying capacity briefly late in the experiment.

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

CONCEPT CHECK 53.3

1. Explain why a population that fits the logistic growth model increases more rapidly at an intermediate size than at relatively small and large sizes.
2. **WHAT IF?** Given the latitudinal differences in sunlight intensity (see Figure 52.3), how might you expect the carrying capacity of plant species found at the equator to compare with that of plant species found at high latitudes?
3. **MAKE CONNECTIONS** Suppose that a sudden change in environmental conditions caused a substantial drop in a population's carrying capacity. Predict how natural selection and genetic drift might affect this population. (See Concept 23.3.)

For suggested answers, see Appendix A.

CONCEPT 53.4

Life history traits are products of natural selection

EVOLUTION Natural selection favors traits that improve an organism's chances of survival and reproductive success. In every species, there are trade-offs between survival and reproductive traits such as frequency of reproduction, number of offspring (number of seeds produced by plants; litter or clutch size for animals), or investment in parental care. The traits that affect an organism's schedule of reproduction and survival make up its **life history**. Life history traits of an organism are evolutionary outcomes reflected in its development, physiology, and behavior.

Diversity of Life Histories

We'll focus on three key components of an organism's life history: when reproduction begins (the age at first reproduction or age at maturity), how often the organism reproduces, and how many offspring are produced per reproductive episode. The fundamental idea that evolution accounts for the diversity of life is manifest in the broad range of these life history characteristics in nature. For example, the age at which reproduction begins varies considerably across species. A typical loggerhead turtle is about 30 years old when it first crawls onto a beach to lay eggs. In contrast, the coho salmon (*Oncorhynchus kisutch*) is often only three or four years old when it spawns.

Organisms also vary in how often they reproduce. The coho salmon is an example of organisms that undergo a "one-shot" pattern of big-bang reproduction, or **semelparity** (from the Latin *semel*, once, and *parere*, to beget). It hatches in the headwaters of a freshwater stream and then migrates to the Pacific Ocean, where it typically requires a few years to mature. The salmon eventually returns to the same stream to spawn, producing thousands of eggs in a single reproductive opportunity before it dies. Semelparity also occurs in some plants, such as the agave, or "century plant" (Figure 53.12a). Agaves generally grow in arid climates with unpredictable rainfall and poor soils. An agave grows for years, accumulating nutrients in its tissues, until there is an unusually wet year. It then sends up a large flowering stalk, produces seeds, and dies. This life history appears to be an adaptation to the agave's harsh desert environment.